



Django startapp Command & App Architecture

Lecture #6: Project vs App • python manage.py startapp • App Structure • INSTALLED_APPS Register

Master Class Notes • Mohit Decodes Hindi Series #6 • Easy Hinglish

🎯 Welcome Soubhik! (Modular Architecture & Database Synergy):

Pichle lecture me humne project structure dekha tha. Ab hum Django ka sabse powerful concept seekhenge — **Apps!** Django ka rule hai: *"Project is the entire house, and Apps are individual functional rooms."* Aapke Python logic aur SQL database tables aage jakar isi App ke andar `models.py` aur `views.py` me likhe jayenge!

🏗️ 1. Project vs App — Dono Me Asli Farq Kya Hai?

🏠 Django Project (Container)

- Pure web application ka **overall container** hota hai.
- Isme global settings (`settings.py`) aur top-level routing (`urls.py`) hoti hai.
- Ek Project ke andar **multiple Apps** ho sakti hain.

🌱 Django App (Independent Module)

- Ek chhota, standalone aur reusable component hota hai jo ek specific kaam karta hai.
- Example: `blog` app, `payment` app, `users_auth` app.
- Ek App ko ek project se nikaal kar kisi **dusre project** me bhi use kiya ja sakta hai!

🍕 Asli Zindagi Ka Example (E-Commerce / Lighting Store):

- **Project:** Aapki poori e-commerce website (e.g., *MyLightingStore*).
- **App 1 (Products App):** Brass lights, pendants, chandeliers display aur filter karne ke liye.
- **App 2 (Cart & Orders App):** Shopping cart aur checkout manage karne ke liye.
- **App 3 (Payment App):** UPI/Razorpay payment gateway handle karne ke liye.
- **Fayda:** Agar kal aapko naya business shuru karna ho, toh wahi *Payment App* bina dobara code likhe naye project me plug-in ho jayegi!

⚙️ 2. Nayi App Create Karne Ka Command (Step-by-Step)



Step 1: Outer Folder me jaayein jahan `manage.py` hai

```
cd MyProject
```

Step 2: startapp Command Chalayein

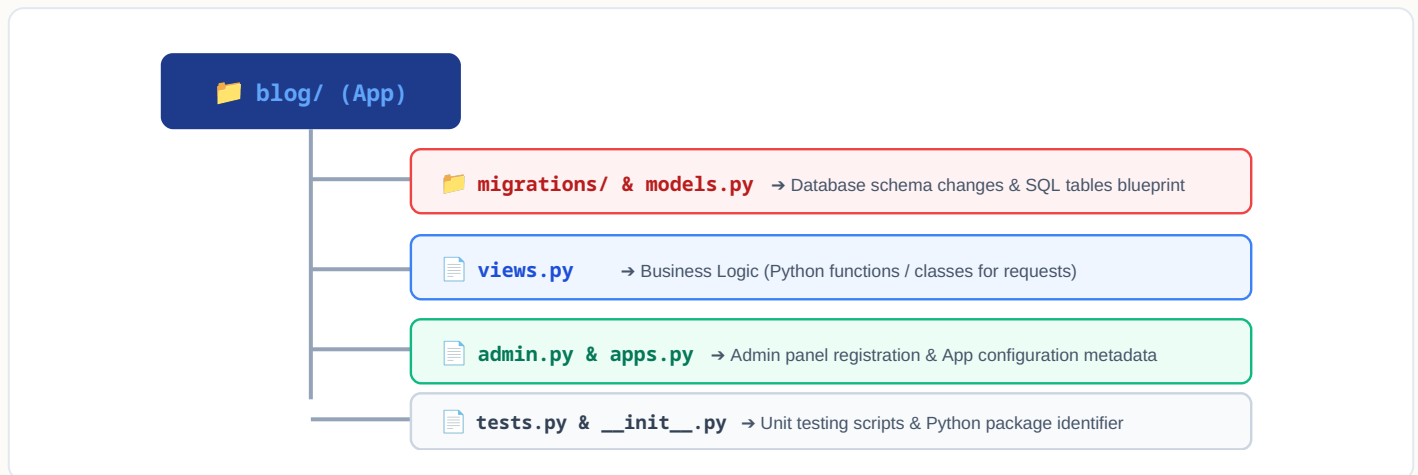
```
# Windows (CMD / PowerShell):  
python manage.py startapp blog
```

```
# macOS / Linux:  
python3 manage.py startapp blog
```

Enter press karte hi `manage.py` ke barabar me ek naya `blog/` naam ka folder ban jayega.

3. App Folder Structure (Har Ek File Ka Role)

Nayi App create hote hi uske andar 6 zaroori files aur ek `migrations/` folder generate hota hai:



File / Directory	Kaam aur Asli Role (Simple Hinglish)
<code>migrations/</code>	Database ke schema changes (jaise naya column add hua ya table bani) ka track record yahan store hota hai. Iske andar <code>__init__.py</code> rehti hai.
<code>models.py</code>	Database Blueprint! Yahan aap Python classes ke form me database tables aur fields define karte hain (SQL Tables).
<code>views.py</code>	Business Logic! Yahan aap Python functions ya classes likhte hain jo user ki request accept karke response wapas bhejte hain.
<code>admin.py</code>	Apne <code>models.py</code> ki tables ko Django ke built-in Admin Panel par display aur manage karne ke liye yahan register karte hain.
<code>apps.py</code>	Is app ki specific configuration aur metadata (jaise app ka naam) define karta hai.
<code>tests.py</code>	App ke code ki automated Unit Testing likhne ke liye use hota hai.

4. App Ko Project Me Register Karna (Sabse Crucial Step)

App bana lena hi kaafi nahi hai. Django ko tab tak is app ke baare me pata nahi chalega jab tak aap isse project ki `settings.py` me register nahi karenge!

File open karein: `MyProject/settings.py` aur `INSTALLED_APPS` list ke andar apni app ka naam add karein:

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    # Hamari nayi App yahan register karein:
    'blog',
]
```

Test Setup (Server Run Karke Verify Karein):

App register karne ke baad server chalayein:

```
python manage.py runserver
```

Agar terminal me koi red error nahi aaya aur **Quit the server with CONTROL-C** dikh gaya, iska matlab aapki **blog app perfectly registered aur linked** ho chuki hai!

5. Top 5 Interview / Viva Quick Revision Q&A

1. **Q: Django Project aur Django App me kya antar hai?**

Ans: Project ek overall container hota hai jisme configuration hoti hai; jabki App ek independent aur reusable module hota hai jo specific functionality (e.g. blog, auth, cart) deliver karta hai.

2. **Q: Nayi App create karne ka CLI command kya hota hai?**

Ans: `python manage.py startapp <app_name>` (macOS: `python3 manage.py startapp <app_name>`).

3. **Q: App create karne ke baad use kahan register karna zaroori hota hai?**

Ans: Project ke `settings.py` file ke andar `INSTALLED_APPS` list me app ka naam add karna hota hai.

4. **Q: `models.py` aur `admin.py` ka aapas me kya rishta hai?**

Ans: `models.py` me database tables banayi jaati hain aur `admin.py` me un tables ko Django Admin Dashboard par manage karne ke liye register kiya jata hai.

5. **Q: Kya ek Django App ko kisi dusre Project me reuse kiya ja sakta hai?**

Ans: Haan, Django apps modular aur pluggable hoti hain, isliye ek app ko dusre project me copy karke `settings.py` me register karke seedha use kiya ja sakta hai.